

elements_view needs its own sentinel

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LWG
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§ 1 Abstract

`elements_view` needs its own `sentinel` type. This paper resolves [\[LWG3386\]](#).

§ 2 Revision history

— R1: Incorporate LWG review feedback in Prague. Rebased onto [\[N4849\]](#).

§ 3 Discussion

`elements_view` is effectively a specialized version of `transform_view`; the latter has a custom `sentinel` type, and so should `elements_view`.

More generally, if a range adaptor has a custom iterator, it should probably also have a custom sentinel. The rationale is that the underlying range's sentinel could encode a generic predicate that is equally meaningful for the adapted range, leading to an ambiguity.

For example, consider:

```
struct S { // sentinel that checks if the second element is zero
    friend bool operator==(input_iterator auto const& i, S) requires /* ... */
    { return get<1>(*i) == 0; }
};

void algo(input_range auto&& r) requires /* ... */ {
    for (auto&& something : subrange{ranges::begin(r), S{}})
    {
        /* do something */
    }
}

using P = pair<pair<char, int>, long>;
vector<P> something = /* ... */;
```

```
subrange r{something.begin(), S{}};

algo(r | view::keys);           // checks the long, effectively iterating over r
                                completely
algo(r | view::transform(&P::first)); // checks the int and stops at the first zero
```

algo is trying to use the sentinel S to stop at the first element e for which `get<1>(e)` is zero, and it works correctly for all ranges of pair<char, int> ... except things like `r | view::keys`. In the latter case, `ranges::begin(r | view::keys) == S{}` calls `elements_view::iterator`'s `operator==` instead, which compares the *underlying range*'s iterator against the sentinel. In the above example, this means that we check the `long` instead of the `int`, and effectively iterate over the entire range.

A custom sentinel is needed to avoid this problem. `elements_view` is the only adaptor in the current WP that has a custom iterator but not a custom sentinel.

§ 4 Wording

This wording is relative to [\[N4849\]](#).

[Drafting note: These changes, including the overload set for `end`, match the specification of `transform_view`. This seems appropriate since `elements_view` is just a special kind of `transform`.]

Edit 24.7.15.2 [\[range.elements.view\]](#), class template `elements_view` synopsis, as indicated:

```
namespace std::ranges {
    [...]

    template<input_range V, size_t N>
        requires view<V> && has_tuple_element<range_value_t<V>, N> &&
            has_tuple_element<remove_reference_t<range_reference_t<V>>, N>
        class elements_view : public view_interface<elements_view<V, N>> {
        public:
            [...]

+         constexpr auto end()
+         { return sentinel<false>{ranges::end(base_)}; }
+
+         constexpr auto end() requires common_range<V>
+         { return iterator<false>{ranges::end(base_)}; }
+
+         constexpr auto end() const
+         requires range<const V>
+         { return sentinel<true>{ranges::end(base_)}; }
+
+         constexpr auto end() const
+         requires common_range<const V>
+         { return iterator<true>{ranges::end(base_)}; }

-         constexpr auto end() requires (!simple_view<V>)
-         { return ranges::end(base_); }

-         constexpr auto end() const requires simple_view<V>
```

```

- { return ranges::end(base_); }

[...]
```

```

private:
    template<bool> struct iterator; // exposition only
+ template<bool> struct sentinel; // exposition only
    V base_ = V(); // exposition only
};
}

```

Edit 24.7.15.3 [\[range.elements.iterator\]](#), class template `elements_view<V, N>::iterator` synopsis, as indicated:

```

namespace std::ranges {
    template<class V, size_t N>
    template<bool Const>
    class elements_view<V, N>::iterator { // exposition only
    [...]

    public:
        [...]

-     friend constexpr bool operator==(const iterator& x, const
        sentinel_t<base_t>& y);

        [...]

-     friend constexpr difference_type
-     operator-(const iterator<Const>& x, const sentinel_t<base_t>& y)
-     requires sized_sentinel_for<sentinel_t<base_t>, iterator_t<base-
        t>>;
-     friend constexpr difference_type
-     operator-(const iterator<base_t>& x, const iterator<Const>& y)
-     requires sized_sentinel_for<sentinel_t<base_t>, iterator_t<base-
        t>>;
    };
}

```

Delete the specification of these operators in 24.7.15.3 [\[range.elements.iterator\]](#) (p13, 23 and 24):

```

friend constexpr bool operator==(const iterator& x, const sentinel_t<base-
    t>& y);

```

13 ~~Effects: Equivalent to: `return x.current_ == y;`~~

...

```

friend constexpr difference_type
operator-(const iterator<Const>& x, const sentinel_t<base_t>& y)
requires sized_sentinel_for<sentinel_t<base_t>, iterator_t<base_t>>;

```

23 ~~Effects: Equivalent to: `return x.current_ == y;`~~

```
friend constexpr difference_type
operator (const sentinel_t<base_t>& x, const iterator<Const>& y)
requires sized_sentinel_for<sentinel_t<base_t>, iterator_t<base_t>>;
```

24 *Effects:* Equivalent to: ~~return (y == x);~~

Add a new subclause after 24.7.15.3 [\[range.elements.iterator\]](#):

§ ??? Class template `elements_view::sentinel` [\[ranges.elements.sentinel\]](#)

```
namespace std::ranges {
    template<class V, size_t N>
    template<bool Const>
    class elements_view<V, N>::sentinel {           // exposition only
    private:
        using Base = conditional_t<Const, const V, V>; // exposition only
        sentinel_t<Base> end_ = sentinel_t<Base>(); // exposition only
    public:
        sentinel() = default;
        constexpr explicit sentinel(sentinel_t<Base> end);
        constexpr sentinel(sentinel<!Const> other)
            requires Const && convertible_to<sentinel_t<V>, sentinel_t<Base>>;

        constexpr sentinel_t<Base> base() const;

        friend constexpr bool operator==(const iterator<Const>& x, const sentinel& y);

        friend constexpr range_difference_t<Base>
            operator-(const iterator<Const>& x, const sentinel& y)
                requires sized_sentinel_for<sentinel_t<Base>, iterator_t<Base>>;

        friend constexpr range_difference_t<Base>
            operator-(const sentinel& x, const iterator<Const>& y)
                requires sized_sentinel_for<sentinel_t<Base>, iterator_t<Base>>;
    };
}
```

1 *Effects:* Initializes `end_` with `end`.

```
constexpr sentinel(sentinel<!Const> other)
    requires Const && convertible_to<sentinel_t<V>, sentinel_t<Base>>;
```

2 *Effects:* Initializes `end_` with `std::move(other.end_)`.

```
constexpr sentinel_t<Base> base() const;
```

3 *Effects:* Equivalent to: `return end_;`

```
friend constexpr bool operator==(const iterator<Const>& x, const sentinel&
    y);
```

4 *Effects:* Equivalent to: `return x.current_ == y.end_;`

```
friend constexpr range_difference_t<Base>
operator-(const iterator<Const>& x, const sentinel& y)
    requires sized_sentinel_for<sentinel_t<Base>, iterator_t<Base>>;
```

5 *Effects:* Equivalent to: return `x.current_ - y.end_;`

```
friend constexpr range_difference_t<Base>
operator-(const sentinel& x, const iterator<Const>& y)
    requires sized_sentinel_for<sentinel_t<Base>, iterator_t<Base>>;
```

6 *Effects:* Equivalent to: return `x.end_ - y.current_;`

§ 5 References

[LWG3386] Tim Song. `elements_view` needs its own sentinel type.

<https://wg21.link/lwg3386>

[N4849] Richard Smith. 2020. Working Draft, Standard for Programming Language C++.

<https://wg21.link/n4849>